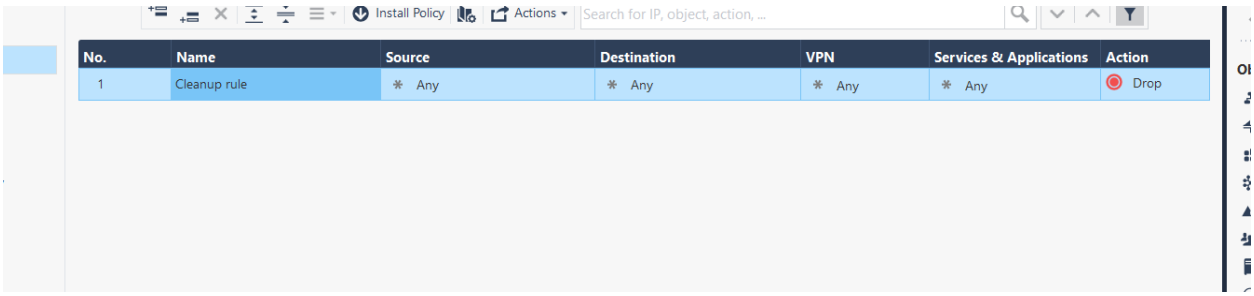
CCSA104 - Practical Assignment: Managing Security Policies

Course: CCSA104 - Security Policy Management
Student Name: Irondi Ugochukwu
Student ID: C11/26/CCSA/17080
Date: 5 November 2026
Score: ____ / 100

Part A: Practical Tasks (70 Marks)

Task 1: Explore the Default Policy (5 Marks)

Policy package name:



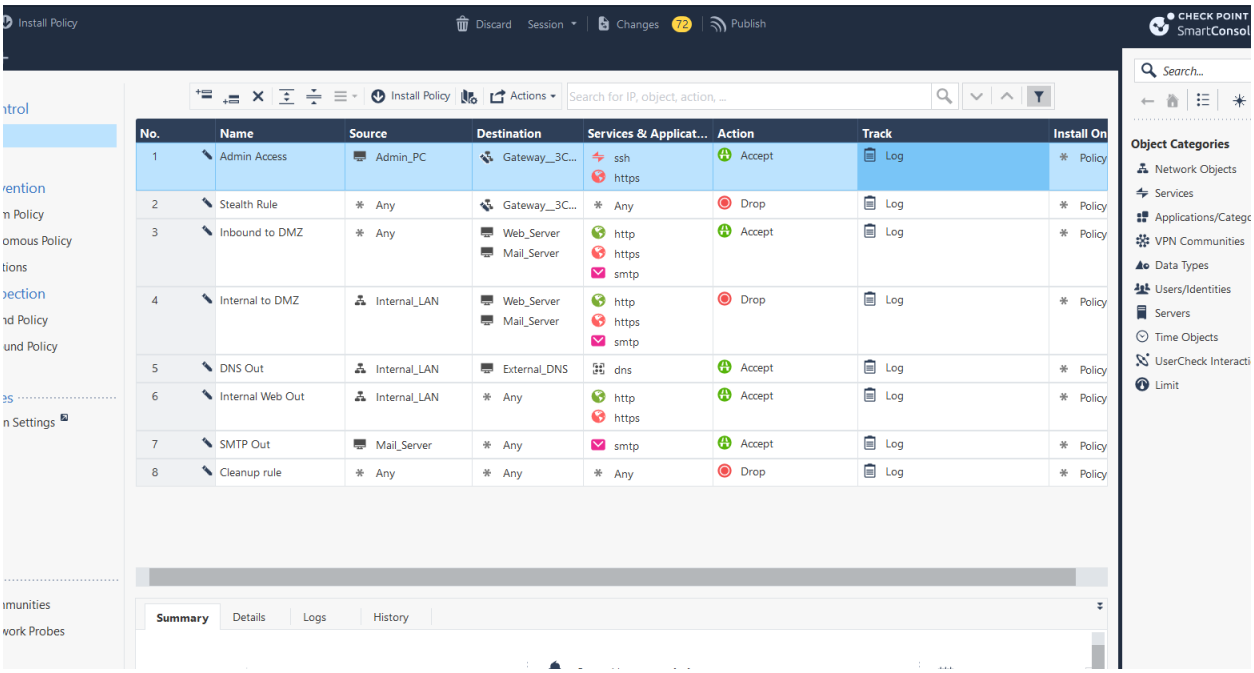
Default: Cleanup Rule

Task 2: Build the Access Control Rulebase (20 Marks)

The following rules define the required access control policy for Orion Logistics. The existing Cleanup Rule is retained at the bottom and its tracking action is set to Log.

No.	Rule Name	Source	Destination	Services	Action	Track
1	Admin Access	Admin_PC	Gateway	SSH, HTTPS	Accept	Log
2	Stealth Rule	Any	Gateway	Any	Drop	Log
3	Inbound to DMZ	Any	Web_Server, Mail_Server	HTTP, HTTPS, SMTP	Accept	Log
4	Internal to DMZ	Internal_LAN	DMZ_Net	HTTP, HTTPS	Drop	Log
5	DNS Out	Internal_LAN	External_DNS	DNS (UDP/TCP)	Accept	Log
6	Internal Web Out	Internal_LAN	Any	HTTP, HTTPS	Accept	Log
7	SMTP Out	Mail_Server	Any	SMTP	Accept	Log
8	Cleanup Rule	Any	Any	Any	Drop	Log

Evidence:



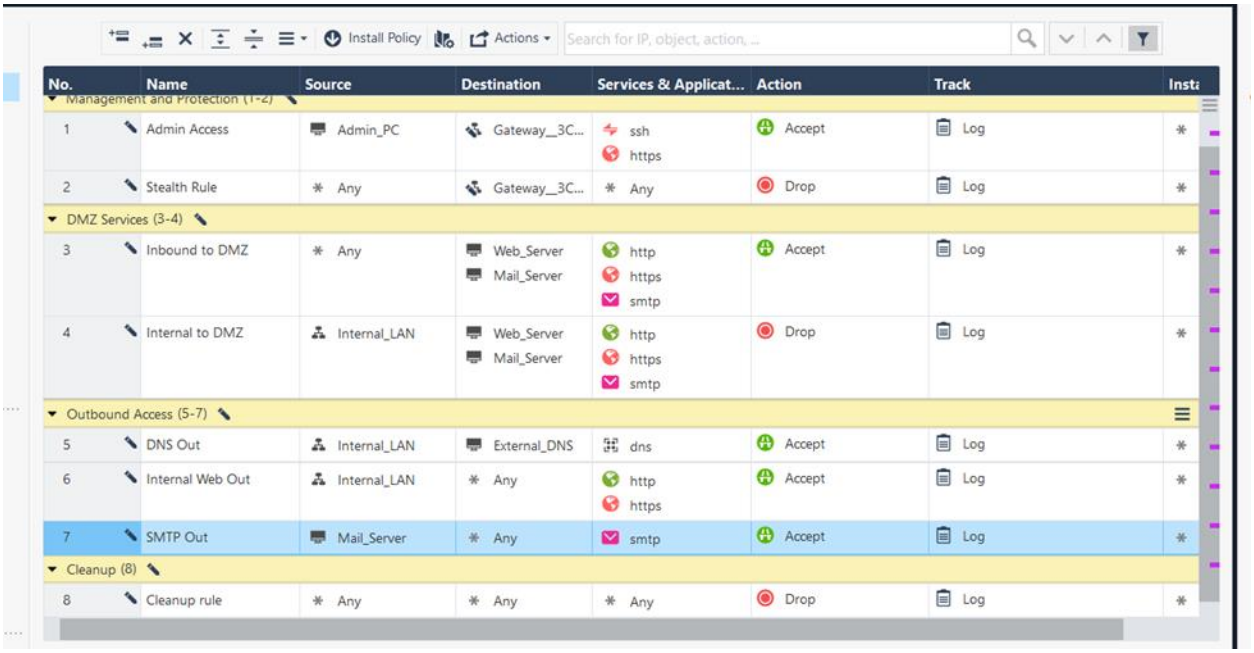
Task 4: Organise the Rulebase with Sections (5 Marks)

The rules are organised into four sections:

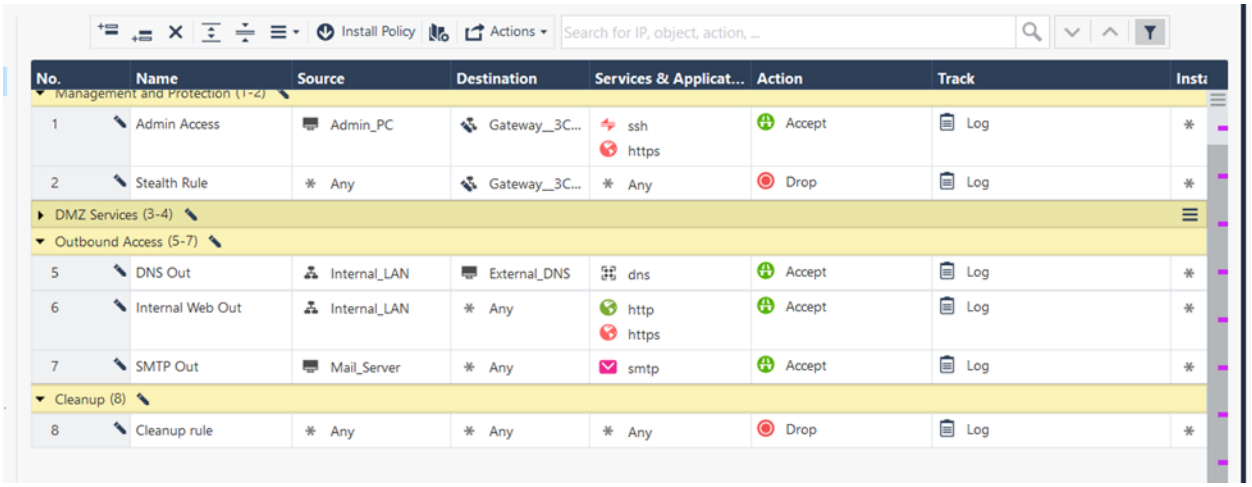
Section	Rules
Management and Protection	Admin Access, Stealth Rule
DMZ Services	Inbound to DMZ, Internal to DMZ
Outbound Access	DNS Out, Internal Web Out, SMTP Out
Cleanup	Cleanup Rule

Observation: Sections are visual organisational elements. They do not change rule enforcement order and are not installed as policy rules on the Security Gateway.

Evidence:



Sections expanded.



One section collapsed.

Task 6: Configure Hide NAT (10 Marks)

NAT Configuration

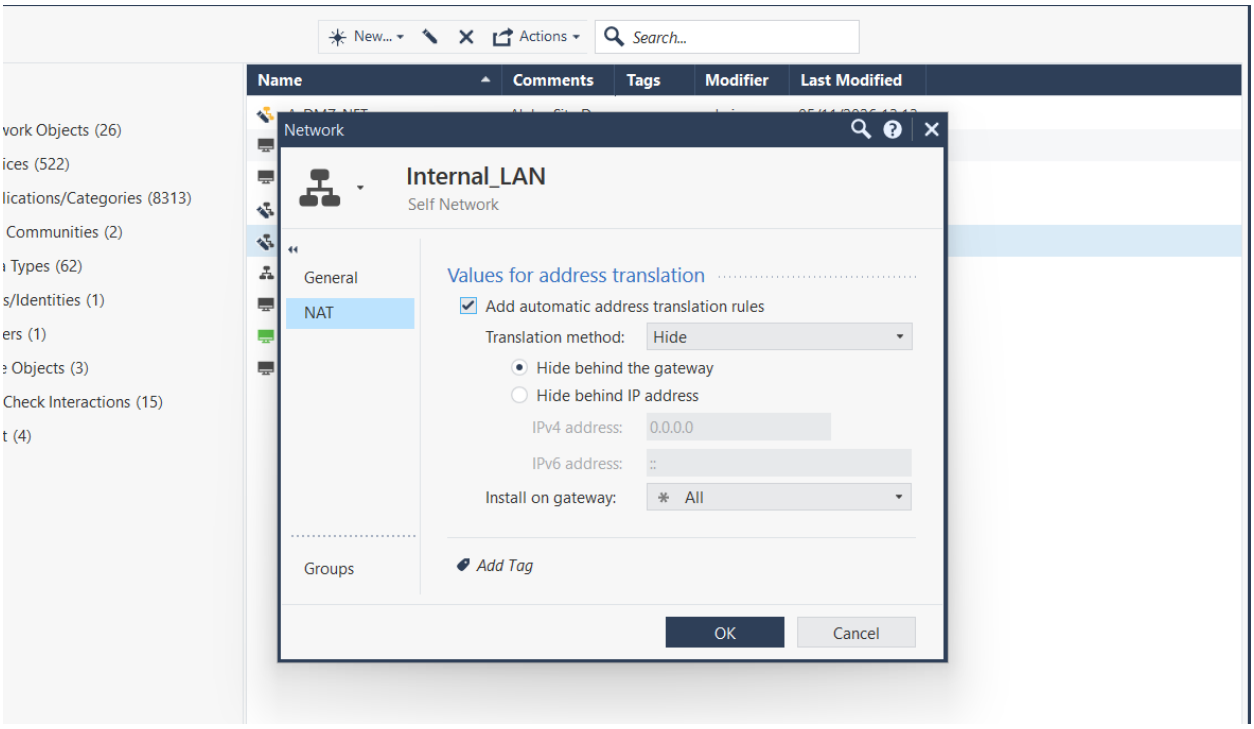
- Object: Internal_LAN
- Translation method: Hide
- Hide behind: Gateway
- NAT type: Automatic

Explanation

Automatic NAT is created from the NAT settings configured directly on the network object. Manual NAT requires the administrator to create and define the translation rule separately.

Hide NAT allows multiple internal IP addresses to share one translated public IP address when accessing external networks. The gateway uses different source port numbers to distinguish individual connections and maintain the correct return traffic mapping.

Evidence:



Part B: Theory Questions (30 Marks)

Section B1: Multiple Choice (5 Marks)

Question	Correct answer
Q1	B -Created in SmartConsole, stored on the Security Management Server and enforced by Security Gateways.
Q2	A - Positive Control Model (whitelist).
Q3	C - Immediately after the Management rules, as early as possible.
Q4	D - Sections are visual divisions and do not affect enforcement order.
Q5	B - Many-to-one translation.

Section B2: Short Answers (15 Marks)

Q6. Distinguish between Explicit rules and Implied rules. State where Implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?

Explicit rules are created by administrators in the Access Control rulebase and are visible as normal policy rules. Implied rules are automatically generated by Check Point to support essential functions such as management communication and other system services. They are configured under Global Properties, and their placement can be controlled relative to the explicit rulebase. The Implicit Cleanup Action determines what happens to traffic that does not match an explicit or implied rule; it is normally dropped.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.

Rules are evaluated from top to bottom, and the first matching rule determines the action. For example, Admin Access must appear before the Stealth Rule so that authorised SSH and HTTPS management traffic can reach the gateway before other traffic to the gateway is dropped.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organisations publish only during a change window?

Saving retains changes in the administrator's working session, while publishing commits those changes to the Security Management Server database so they become part of the shared management configuration. Unpublished changes are not included when a policy is installed. Organisations publish during approved change windows to control when configuration changes become available, reduce unexpected changes and ensure that changes are reviewed before deployment.

Q9. Compare Global Properties with Policy Packages. Give one example of what each controls and say how you open each in SmartConsole.

Global Properties contain management-wide settings that can affect multiple policies, such as implied rule behaviour. They are accessed through Menu/Application menu → Global Properties. Policy Packages contain the specific Access Control and other policy layers assigned for enforcement on selected gateways. They are accessed through Manage Policies and Layers in SmartConsole.

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone-based rules.

The four predefined zones are:

1. Internal
2. External
3. DMZ
4. Undefined

A gateway interface can belong to only one security zone, and the zone must be assigned to the correct interface based on the network it connects to. Zone-based rules simplify policy management because administrators can control traffic between network areas without listing every individual network object.

Section B3: Scenario Questions (10 Marks)

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services Any and Action Accept above the Stealth rule. Describe two security problems this causes, explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

The rule creates unrestricted access because it allows traffic from any source to any destination using any service. It also makes the Stealth Rule ineffective because traffic matching the unrestricted rule is accepted before reaching the Stealth Rule.

The administrator should remove the unrestricted rule and replace it with specific rules that allow only required traffic. The only rule expected above the Stealth Rule is the Management Rule, which permits authorised administrators and management systems to communicate with the gateway.

Q12. After Hide NAT is configured for Internal_LAN, staff report they cannot reach the Internet.

(a) State two benefits of NAT.

1. It allows multiple private internal devices to access the Internet using a shared public IP address.
2. It hides internal IP addressing from external networks, reducing direct exposure of the internal addressing structure.

(b) Describe three checks you would make to find the fault, and explain why each one matters.

1. **Check the NAT rule:** Confirm that the automatic Hide NAT rule exists, is enabled and translates Internal_LAN behind the correct gateway address. This verifies that internal traffic is being translated.
2. **Check the Access Control policy:** Confirm that the relevant outbound traffic is allowed and that the policy has been installed on the gateway. NAT does not automatically permit traffic through the firewall.
3. **Check routing and connectivity:** Verify that the gateway has a valid default route and can reach the external network. Without a working route, translated traffic cannot reach the Internet.